

Satellite Project: Cloud Masking

Team Members

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Project Pipeline

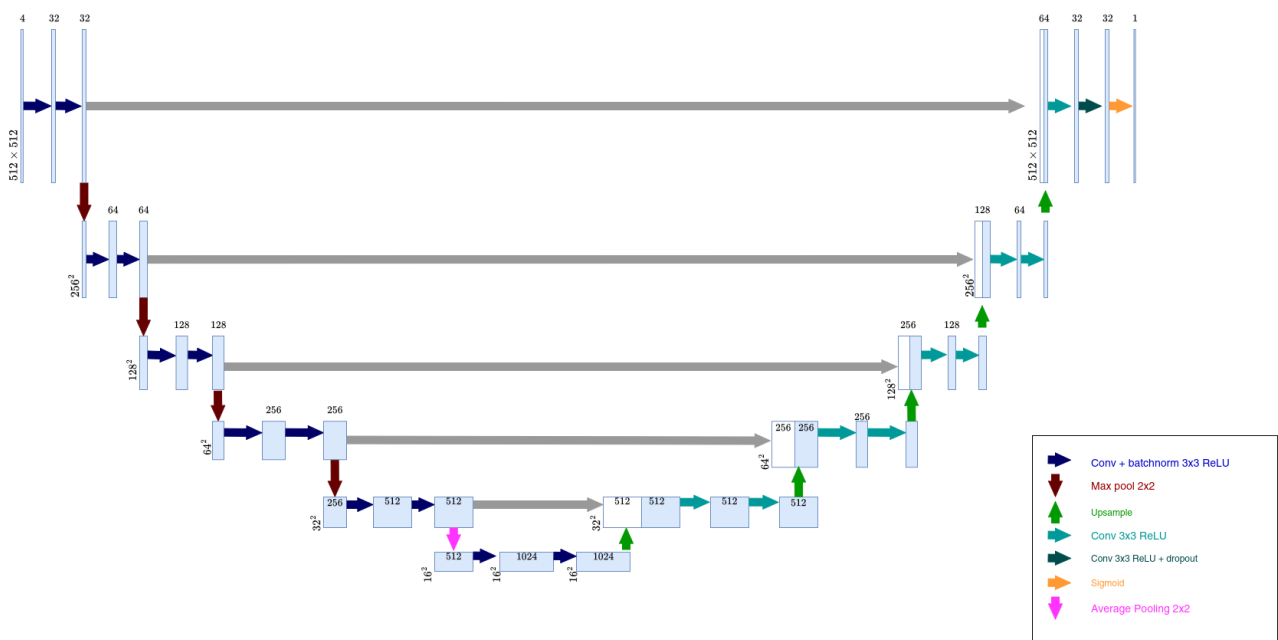


Figure 1: Project Architecture

PreProcessing

The preprocessing pipeline includes:

- Loading and organizing satellite images and their corresponding masks
- Converting GDAL datasets to NumPy arrays for processing
- Splitting the dataset into training (80%), validation (20%), and test (20%) sets.
- Applying random horizontal and vertical flips and random rotations to augment the dataset.

Exploratory Data Analysis

- Dataset consists of 10,573 satellite images with corresponding cloud masks
- Images are in TIFF format with 4 spectral bands
- Masks are binary (cloud vs. no-cloud)
- Image dimensions: 512x512 pixels
- Data distribution:
 1. Training: 6,343 images
 2. Validation: 2,115 images
 3. Test: 2,115 images

Model Selection and Training Module

Model Architecture

We implemented a U-Net architecture with:

- Encoder-decoder structure with skip connections
- 5 levels of downsampling and upsampling
- Batch normalization and dropout for regularization
- Final sigmoid activation for binary classification

Training Parameters

- Batch size: 16
- Learning rate: 0.001
- Number of epochs: 10
- Loss function: Cross Entropy Loss
- Optimizer: SGD
- Dropout probability: 0.1

Performance Analysis Module

Evaluation Metrics

- Dice Coefficient: 0.8849 on test set
- Model has 7,764,353 parameters
- Approximately 1.48 trillion operations per forward pass

Training Results

The model showed consistent improvement during training:

- Stable decrease in training loss
- Good generalization on validation set
- Final test performance indicates effective learning

Enhancement and Future Work

Potential improvements:

- Experiment with different loss functions (Dice loss, Focal loss)
- Try more advanced architectures (Attention U-Net, ResUNet)
- Implement data augmentation for better generalization
- Optimize hyperparameters using grid search

- Explore transfer learning from pre-trained models
- Implement model quantization for deployment